CSAI2121

Probability and Statistics

Integrated Exercise Set

Chapters 5–7

Joint Distributions, Covariance & the Multivariate Gaussian,
the Law of Large Numbers & the Central Limit Theorem

Total: 24 Exercises

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Part I Joint Distributions of Random Variables

Exercise 1: A Joint PMF for Cloud Deployment Decisions

A machine learning platform logs two attributes of each deployment job: X = deployment region (0 =Asia, 1 =Europe) and Y = observed latency category (0 =Low, 1 =Medium, 2 =High). The joint PMF is

	$Y=0$	$Y=1$	$Y=2$
$X=0$	0.10	0.25	0.05
$X=1$	0.05	0.20	0.35

(a) Find the marginal PMF of X and the marginal PMF of Y.

(b) Find $P(Y=2 \mid X=1)$.

(c) Find $P(X=0 \mid Y=0)$.

(d) Determine whether X and Y are independent. Justify your answer using the definition from Chapter 5.

(e) Find $P(X=1,\ Y \geq 1)$.

(f) Find the conditional PMF of X given $Y=1$, and verify that it sums to 1.

Exercise 2: From a Fraud-Detection Confusion Matrix to a Joint PMF

A fraud-detection system is evaluated on $n=500$ transactions. Let X denote the model's prediction (1 =flagged, 0 =not flagged) and Y denote the true status (1 =fraud, 0 =legitimate). The counts are

	$Y=1$ (fraud)	$Y=0$ (legitimate)
$X=1$ (flagged)	40	60
$X=0$ (not flagged)	10	390

(a) Convert the table into an empirical joint PMF $p_{X,Y}(x,y)$ by dividing each count by n.

(b) Find the marginal PMF of X and of Y.

(c) Find the precision $P(Y=1 \mid X=1)$.

(d) Find the recall $P(X=1 \mid Y=1)$.

(e) Determine whether X and Y are independent.

(f) Find $P(X=0 \mid Y=1)$ and verify that it equals $1-$ recall.

Exercise 3: A Separable Continuous Joint Density

A continuous joint PDF is given by

$$f_{X,Y}(x,y) = \begin{cases} c\,xy^2, & 0 \le x \le 1,\ 0 \le y \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of c using an iterated integral.

(b) Find the marginal PDF $f_X(x)$.

(c) Find the marginal PDF $f_Y(y)$.

(d) Find $P(X < 0.5,\ Y < 1)$.

(e) Determine whether X and Y are independent.

(f) Find $E[X]$, $E[Y]$, and $E[XY]$, and verify that $E[XY] = E[X]E[Y]$.

Exercise 4: A Non-Separable Continuous Joint Density

A continuous joint PDF is given by

$$f_{X,Y}(x,y) = \begin{cases} k(2x+y), & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of k.

(b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

(c) Find $P(X > 0.5,\ Y > 0.5)$.

(d) Determine whether X and Y are independent by checking $f_X(x)f_Y(y)$ against $f_{X,Y}(x,y)$ at $(0,0)$.

(e) Find the conditional PDF $f_{X|Y}(x \mid y = 0.5)$, and compare it with $f_X(x)$.

(f) Find the conditional PDF $f_{X|Y}(x \mid y = 0)$, and explain why parts (e) and (f) together confirm the conclusion of part (d).

Exercise 5: Independent Sensor Readings and Anomaly Thresholds

Two independent exponential sensors measure temperature deviation X (rate $\lambda_1 = 0.5$) and vibration Y (rate $\lambda_2 = 0.25$), both taking values on $[0, \infty)$. The joint PDF is

$$f_{X,Y}(x,y) = 0.125\, e^{-0.5x - 0.25y}, \qquad x, y \ge 0.$$

(a) Verify that $f_{X,Y}$ integrates to 1.

(b) Find the marginal distributions of X and Y, and identify their type.

(c) Show that X and Y are independent.

(d) An anomaly is flagged if $X > 4$ or $Y > 8$. Find the probability of *no* anomaly, and hence the probability of an anomaly.

(e) Find $P(X > 4,\ Y > 8)$.

(f) Discuss why independence made the computations in (d)–(e) straightforward, and what would change if X and Y were correlated.

Exercise 6: The Naive Bayes Independence Assumption in Spam Filtering

A spam filter uses two binary features: $X_1 = 1$ if the message contains the word "limited", and $X_2 = 1$ if it contains the word "offer". The class-conditional probabilities are

$$P(X_1 = 1 \mid \text{spam}) = 0.60, \quad P(X_2 = 1 \mid \text{spam}) = 0.45,$$
$$P(X_1 = 1 \mid \text{ham}) = 0.05, \quad P(X_2 = 1 \mid \text{ham}) = 0.02,$$

with prior $P(\text{spam}) = 0.30$. Naive Bayes assumes X_1, X_2 are conditionally independent given the class.

(a) Under the Naive Bayes assumption, find $P(X_1 = 1, X_2 = 1 \mid \text{spam})$ and $P(X_1 = 1, X_2 = 1 \mid \text{ham})$.

(b) Using Bayes' theorem, find $P(\text{spam} \mid X_1 = 1, X_2 = 1)$.

(c) Suppose the *true* joint probability is actually $P(X_1 = 1, X_2 = 1 \mid \text{spam}) = 0.33$ (rather than the value implied by independence). Recompute the posterior probability using this true value.

(d) Compare the two posterior probabilities from parts (b) and (c).

(e) Discuss why Naive Bayes classifiers can still perform well in practice even when the independence assumption is not exactly correct.

Part II Covariance, Correlation, and the Multivariate Gaussian

Exercise 7: Covariance and Correlation from a Joint PMF

Let X denote model complexity (0 =simple, 1 =moderate, 2 =complex) and Y denote training time (0 =short, 1 =long), with joint PMF

	$Y=0$	$Y=1$
$X=0$	0.30	0.05
$X=1$	0.15	0.15
$X=2$	0.05	0.30

(a) Find the marginal PMFs of X and Y.

(b) Find $E[X]$ and $E[Y]$.

(c) Find $E[XY]$.

(d) Find $\mathrm{Cov}(X, Y)$.

(e) Find $\mathrm{Var}(X)$, $\mathrm{Var}(Y)$, and the correlation coefficient $\rho_{X,Y}$.

(f) Interpret the sign and approximate magnitude of $\rho_{X,Y}$ in the context of model complexity and training time.

Exercise 8: Algebraic Properties of Covariance and Variance

Suppose $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = -3$.

(a) Find $\mathrm{Var}(X + Y)$.

(b) Find $\mathrm{Var}(X - Y)$.

(c) Find $\mathrm{Var}(2X + 3Y)$.

(d) Find $\mathrm{Cov}(X + 2,\ 3Y - 5)$.

(e) Find $\mathrm{Cov}(2X,\ -Y)$.

(f) Find the correlation coefficient $\rho_{X,Y}$.

(g) Find $\mathrm{Cov}(X + Y,\ X - Y)$ in terms of $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$, and comment on whether $X + Y$ and $X - Y$ are correlated here.

Exercise 9: Zero Correlation Without Independence

Let $X \sim \mathrm{Uniform}(-3, 3)$ and define $Y = X^2$.

(a) Find $E[X]$ and $\mathrm{Var}(X)$.

(b) Find $E[Y] = E[X^2]$.

(c) Find $E[XY] = E[X^3]$, and explain why this is 0 using symmetry.

(d) Find $\mathrm{Cov}(X, Y)$.

(e) Explain why X and Y are clearly *dependent*, even though their covariance is 0.

(f) Relate this example to the general warning given in Chapter 6 about what zero correlation does and does not imply.

Exercise 10: Constructing and Interpreting a 3×3 Covariance Matrix

Three correlated performance metrics X, Y, Z have

$$\mathrm{Var}(X) = 2, \quad \mathrm{Var}(Y) = 5, \quad \mathrm{Var}(Z) = 3,$$
$$\mathrm{Cov}(X, Y) = 1, \quad \mathrm{Cov}(X, Z) = -1, \quad \mathrm{Cov}(Y, Z) = 0.$$

(a) Write the full 3×3 covariance matrix Σ.

(b) Verify that Σ is symmetric.

(c) Find $\mathrm{Var}(X + Y + Z)$ using the entries of Σ.

(d) Find $\mathrm{Cov}(X + Y,\ Z)$.

(e) Find $\mathrm{Cov}(aX + bY + cZ,\ X)$ in terms of a, b, c.

(f) Does $\mathrm{Cov}(Y, Z) = 0$ necessarily imply that Y and Z are independent? Explain, referring to Exercise 9.

Exercise 11: Shape of the Multivariate Gaussian: Diagonal vs. Full Covariance

Let $(X, Y) \sim N(\boldsymbol{\mu}, \Sigma)$ with $\boldsymbol{\mu} = (2, -1)^T$ and $\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$.

(a) Since Σ is diagonal, write $p(x, y)$ as a product of two univariate Gaussian densities, stating the parameters of each.

(b) Determine whether X and Y are independent, and justify your answer.

(c) Describe the shape of the density contours of (X, Y) (circular, axis-aligned ellipse, or tilted ellipse), and explain why.

(d) Now suppose instead $\Sigma' = \begin{bmatrix} 4 & 3 \\ 3 & 9 \end{bmatrix}$. Compute the correlation coefficient implied by Σ'.

(e) Verify that Σ' is a valid covariance matrix by checking that it is symmetric and has strictly positive eigenvalues.

(f) Describe how the contour shape changes compared to part (c).

Exercise 12: Mahalanobis Distance and Anomaly Detection

Normal sensor behaviour is modelled as $(X_1, X_2) \sim N(\mathbf{0}, \Sigma)$ with $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$. Two new readings are observed: $A = (2, 2)$ and $B = (2, -2)$.

(a) Find Σ^{-1}.

(b) Compute the squared Mahalanobis distance $D_M^2(A)$.

(c) Compute the squared Mahalanobis distance $D_M^2(B)$.

(d) Compute the ordinary squared Euclidean distances of A and B from the origin, and compare them.

(e) Explain why A and B have the same Euclidean distance from the origin, yet different Mahalanobis distances. (Hint: consider the directions $(1, 1)$ and $(1, -1)$ relative to Σ.)

(f) If points with $D_M^2 > 3$ are flagged as anomalous, determine which of A, B (if any) would be flagged.

Part III The Law of Large Numbers and the Central Limit Theorem

Exercise 13: Mean and Variance of a Sample Mean

A cloud service's response time has population mean $\mu = 120$ ms and standard deviation $\sigma = 18$ ms. A sample of $n = 81$ requests is monitored.

(a) Find $E[\overline{X}_{81}]$.

(b) Find $\text{Var}(\overline{X}_{81})$.

(c) Find $\text{SE}(\overline{X}_{81})$.

(d) If n is increased to 324, find the new standard error and compare it with part (c) using the square-root rule.

(e) Find the smallest sample size n such that $\text{SE}(\overline{X}_n) \leq 1$ ms.

(f) Does increasing n to infinity make $\overline{X}_n$ exactly equal to μ? Explain precisely what the Law of Large Numbers does and does not guarantee.

Exercise 14: The Law of Large Numbers via Chebyshev's Inequality

In an A/B test, a user's conversion indicator has mean $p = 0.12$ and variance $p(1-p) = 0.1056$. Let $\widehat{p}$ be the sample proportion from n independent users.

(a) For $n = 400$, find $E[\widehat{p}]$ and $\text{Var}(\widehat{p})$.

(b) Using Chebyshev's inequality, find an upper bound on $P(|\widehat{p} - 0.12| \geq 0.05)$.

(c) Repeat part (b) for $n = 1600$.

(d) Explain, using the Law of Large Numbers, why this upper bound tends to 0 as $n \to \infty$.

(e) Explain why the Chebyshev bound is typically a loose (overly conservative) estimate compared to the true probability.

Exercise 15: Applying the Central Limit Theorem

A dataset feature X has population mean $\mu = 50$ and standard deviation $\sigma = 15$. A random sample of size $n = 100$ is drawn.

(a) State the approximate distribution of $\overline{X}_{100}$ according to the CLT.

(b) Find $P(\overline{X}_{100} > 52)$.

(c) Find $P(47 < \overline{X}_{100} < 53)$.

(d) Find the value c such that $P(\overline{X}_{100} > c) = 0.05$.

(e) Did we need to know the shape of the original distribution of X to answer parts (b)–(d)? Explain.

Exercise 16: Comparing Two Models with the CLT

Two recommendation algorithms' average session durations are measured on independent samples: Model A has $n_A = 120$, population mean $\mu_A = 45$ seconds, $\sigma_A = 12$ seconds; Model B has $n_B = 150$, population mean $\mu_B = 42$ seconds, $\sigma_B = 10$ seconds.

(a) Find $E[\overline{A} - \overline{B}]$.

(b) Find $\text{Var}(\overline{A} - \overline{B})$.

(c) Find $\text{SE}(\overline{A} - \overline{B})$.

(d) Using the CLT, find $P(\overline{A} - \overline{B} > 0)$.

(e) Find $P(|(\overline{A} - \overline{B}) - 3| > 2)$.

(f) State the assumptions (independence, sample size) required to apply the CLT in this setting.

Exercise 17: Planning Sample Size via the Standard Error

A company wants to estimate average model inference latency with a standard error of at most 0.5 ms. Pilot data suggest $\sigma \approx 6$ ms.

(a) Find the minimum sample size n required.

(b) If only $n = 100$ requests can be tested, find the resulting standard error.

(c) By what factor must n increase to halve the standard error found in part (b)?

(d) If σ is actually 9 ms, recompute the minimum n needed for $\text{SE} \leq 0.5$ ms.

(e) Discuss the trade-off between measurement cost (larger n) and estimation precision (smaller SE).

Exercise 18: Classification Accuracy as a Sample Mean

A deployed classifier has true accuracy $p = 0.93$. On a test set of $n = 500$ independent examples, let X be the number of correct predictions and $\widehat{p} = X/500$ the sample accuracy.

(a) State the exact distribution of X, and its mean and variance.

(b) Using the CLT, state the approximate distribution of $\widehat{p}$.

(c) Find the standard error of $\widehat{p}$.

(d) Find $P(\widehat{p} < 0.90)$.

(e) Find $P(0.91 < \widehat{p} < 0.95)$.

(f) Explain, referring to both the LLN and the CLT, why larger test sets produce more trustworthy accuracy estimates.

Part IV Integrated Challenge Problems

Challenge 19: Correlated Ensemble Errors and the Limits of Averaging

Two models' prediction errors $\varepsilon_1, \varepsilon_2$ have mean 0, $\mathrm{Var}(\varepsilon_1) = \mathrm{Var}(\varepsilon_2) = \sigma^2 = 4$, and correlation ρ, so that $\mathrm{Cov}(\varepsilon_1, \varepsilon_2) = 4\rho$. Define the ensemble error $\bar{\varepsilon} = (\varepsilon_1 + \varepsilon_2)/2$.

(a) Derive $\mathrm{Var}(\bar{\varepsilon})$ in terms of ρ.

(b) Evaluate $\mathrm{Var}(\bar{\varepsilon})$ for $\rho = 0$, $\rho = 0.5$, and $\rho = 1$.

(c) Using the "averaging reduces variance" intuition from Chapter 7, explain why the variance reduction disappears as $\rho \to 1$.

(d) For n models with common variance σ^2 and common pairwise correlation ρ, it can be shown that

$$\mathrm{Var}(\bar{\varepsilon}_n) = \sigma^2 \left[\frac{1}{n} + \rho \frac{n-1}{n} \right].$$

Verify this formula for $n = 3$ by directly expanding $\mathrm{Var}(\varepsilon_1 + \varepsilon_2 + \varepsilon_3)$.

(e) Find $\lim_{n \to \infty} \mathrm{Var}(\bar{\varepsilon}_n)$ for fixed $\rho > 0$, and interpret the result.

(f) Discuss why this result shows that a naive application of the Law of Large Numbers ("more models always reduce variance to zero") can be misleading for *correlated* ensembles.

Challenge 20: Multivariate Sensors, Mahalanobis Scores, and the CLT

Two correlated sensor deviations follow $(X, Y) \sim N(\mathbf{0}, \Sigma)$ each hour, with $\Sigma = \begin{bmatrix} 9 & 3 \\ 3 & 4 \end{bmatrix}$, independently across hours. Define the hourly anomaly score S as the squared Mahalanobis distance of that hour's reading.

(a) Verify that Σ is a valid (positive definite) covariance matrix.

(b) Find Σ^{-1}, and write the formula for $S = (x, y)\Sigma^{-1}(x, y)^T$.

(c) **(Given fact)** Under this model, S has mean 2 and variance 4 for each hour. If scores are averaged over $n = 100$ independent hours, find $E[\overline{S}_{100}]$ and $\mathrm{Var}(\overline{S}_{100})$.

(d) Using the CLT, approximate $P(\overline{S}_{100} > 2.5)$.

(e) Explain why the CLT allows an approximately Gaussian treatment of $\overline{S}_{100}$, even though a single hourly score S is *not* Gaussian.

(f) Discuss how a sustained elevation of $\overline{S}_{100}$ above 2.5 over many hours could indicate a systematic sensor fault, connecting this to the LLN idea of concentration around a true mean.

Challenge 21: Naive Bayes Assumptions and Accuracy Uncertainty

Network traffic features (packet size X_1, inter-arrival time X_2) are modelled as bivariate Gaussian given class Y:

$$\text{Class } Y=1: \quad \Sigma_1 = \begin{bmatrix} 2500 & 0 \\ 0 & 0.01 \end{bmatrix}, \qquad \text{Class } Y=0: \quad \Sigma_0 = \begin{bmatrix} 1600 & 3 \\ 3 & 0.02 \end{bmatrix}.$$

(a) For class $Y = 1$, explain why the Naive Bayes conditional-independence assumption is exactly satisfied.

(b) For class $Y = 0$, compute the correlation coefficient ρ implied by Σ_0, and verify that Σ_0 is a valid covariance matrix.

(c) Write the Naive Bayes approximation of $p(x_1, x_2 \mid Y = 0)$ in terms of the univariate marginals, and comment on how strongly this approximation violates the true dependence structure, given ρ from part (b).

(d) Suppose this (approximate) classifier has true accuracy $p = 0.88$ on a batch of $n = 200$ independent flows. Using the CLT, approximate the probability that the observed sample accuracy on this batch falls below 0.85.

(e) In 2–3 sentences, explain how parts (a)–(c) (joint/covariance structure) and part (d) (CLT on accuracy) represent two different, complementary roles of probability theory in evaluating an AI system.

Challenge 22: A Weighted Ensemble: Covariance Matrix Meets the CLT

Three models' prediction errors e_1, e_2, e_3 (each with mean 0) have covariance matrix

$$\Sigma = \begin{bmatrix} 1.0 & 0.3 & -0.1 \\ 0.3 & 1.5 & 0.2 \\ -0.1 & 0.2 & 2.0 \end{bmatrix}.$$

A weighted ensemble combines them as $e_w = 0.5e_1 + 0.3e_2 + 0.2e_3$.

(a) Using $\text{Var}(\mathbf{w}^T\mathbf{e}) = \mathbf{w}^T\Sigma\mathbf{w}$ with $\mathbf{w} = (0.5, 0.3, 0.2)^T$, compute $\text{Var}(e_w)$.

(b) Compare $\text{Var}(e_w)$ to $\text{Var}(e_1) = 1.0$. Does the weighted ensemble reduce variance despite positive correlations among the models?

(c) If e_w is computed independently over $n = 50$ evaluation batches, use the CLT to approximate $P(|\overline{e}_{w,50}| > 0.1)$, where $\overline{e}_{w,50}$ is the average of the 50 independent values of e_w.

(d) Explain why the independence assumption required in part (c) (batch-to-batch independence) is conceptually different from the correlation structure described by Σ (model-to-model correlation within a single batch).

Challenge 23: From a Daily Confusion Matrix to CLT-Based Monitoring

A fraud-detection system flags $m = 50$ transactions per day. For each flagged transaction, assume (idealized) that it is truly fraudulent with probability $\pi = 0.65$, independently across transactions. Let $\widehat{X}$ be the day's sample precision.

(a) Find $E[\widehat{X}]$ and $\mathrm{Var}(\widehat{X})$.

(b) Using the CLT, approximate the probability that a single day's precision falls below 0.55 purely due to random fluctuation.

(c) The company flags the model for review whenever a single day's precision falls below 0.55. Over 20 independent operating days, find the expected number of days that trigger such a *false* alert (i.e., even though π is still truly 0.65).

(d) Treating "trigger a false alert" as an approximately Bernoulli event for each day with the probability found in part (b), use a normal approximation to estimate $P(\text{at least 3 of 20 days trigger a false alert})$.

(e) Discuss the conditions normally required for the normal approximation in part (d) to be reasonable (e.g., np and $n(1-p)$ both reasonably large), and whether they are satisfied here.

(f) Summarize how the joint distribution of (prediction, truth) each day (Chapter 5), its reduction to a daily precision statistic (Chapter 6), and the CLT (Chapter 7) together support a monitoring decision.

Challenge 24: A Credit-Scoring Model: Multivariate Gaussian Meets the CLT

A bank computes a risk score from two correlated features: debt ratio X_1 and payment delay count X_2, with $(X_1, X_2) \sim N(\boldsymbol{\mu}, \Sigma)$, $\boldsymbol{\mu} = (0.4, 2)^T$, $\Sigma = \begin{bmatrix} 0.01 & 0.03 \\ 0.03 & 4 \end{bmatrix}$. The risk score is $R = 10X_1 + 2X_2$.

(a) Find $E[R]$.

(b) Find $\mathrm{Var}(R)$.

(c) State the *exact* distribution of R for a single applicant, and explain why no CLT approximation is needed here.

(d) An applicant is declined if $R > c$. Find c such that $P(R > c) = 0.10$.

(e) For a sample of $n = 25$ independent applicants, find the mean and standard error of the sample average score $\overline{R}_{25}$.

(f) Using the CLT, find $P(\overline{R}_{25} > 9)$.

(g) Explain the conceptual difference between part (c), where Gaussianity of R for one applicant follows exactly from the multivariate Gaussian assumption on (X_1, X_2), and part (f), where the approximate Gaussianity of $\overline{R}_{25}$ follows from the CLT regardless of the original feature distributions.

Formula Reference

Joint, marginal, and conditional distributions

$$p_{X,Y}(x,y) = P(X=x, Y=y), \qquad f_{X,Y}(x,y):\ P(A) = \iint_A f_{X,Y}(x,y)\,dx\,dy.$$

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy.$$

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \qquad f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

Independence

$$X, Y \text{ independent} \iff p_{X,Y}(x,y) = p_X(x)p_Y(y) \text{ (or } f_{X,Y} = f_X f_Y) \text{ for all } x, y.$$

Covariance and correlation

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - \mu_X\mu_Y,$$

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}, \qquad -1 \le \rho_{X,Y} \le 1.$$

$$\mathrm{Var}(aX+bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y).$$

Covariance matrix and multivariate Gaussian

$$\Sigma = \mathrm{Cov}(\mathbf{X}) = E\big[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^T\big], \qquad \Sigma = \Sigma^T, \quad \Sigma_{ii} = \mathrm{Var}(X_i).$$

$$\mathrm{Var}(\mathbf{w}^T\mathbf{X}) = \mathbf{w}^T\Sigma\mathbf{w}.$$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right).$$

$$D_M(\mathbf{x})^2 = (\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu}) \quad \text{(Mahalanobis distance squared)}.$$

Sample mean, LLN, and CLT

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad E[\overline{X}_n] = \mu, \qquad \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\overline{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

$$\text{LLN: } \overline{X}_n \longrightarrow \mu \qquad \text{as } n \to \infty.$$

$$\text{CLT: } \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1), \qquad \text{so for large } n,\ \ \overline{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

End of Exercise Set